

**THE UNIQUE ORDINARY ELEMENT OF A
ONE-GENERATED HEYTING ALGEBRA, THE
SUBGROUP LATTICE OF $\mathbb{Z}/12\mathbb{Z}$, AND A
CHARACTERIZATION OF $n = p^2q$**

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ABSTRACT. Call an element of a Heyting algebra *ordinary* (following Citkin) if it is neither regular ($\neg\neg a = a$) nor dense ($\neg a = \perp$). We collect, with machine-checked proofs, a group of results centered on ordinary elements and the Rieger–Nishimura lattice. (1) *Non-degeneracy*: relative to a fixed element a , every non-bottom element of a Heyting algebra occupies exactly one of four lattice regions determined by a , $\neg a$, and $\neg\neg a$, and all four regions are simultaneously inhabited if and only if a is ordinary. (2) *Threshold*: any Heyting algebra containing an ordinary element has at least six elements, and the six elements $\perp, g, \neg g, \neg\neg g, g \vee \neg g, \top$ realize an order-embedding of the six-element one-generated Heyting algebra Z_6 ; the bound generalizes an observation of Citkin from one-generated algebras with ordinary generators to arbitrary Heyting algebras. (3) *Uniqueness*: in any one-generated Heyting algebra of cardinality ≥ 6 — hence in every finite one-generated algebra Z_n ($n \geq 6$) and in the free Heyting algebra on one generator — the generator is the unique ordinary element. The statement appears without proof in a recent preprint of Citkin; the machine-checked proof given here is, to our knowledge and his, the first written proof. En route we give a machine-checked proof of Nishimura’s one-variable normal-form theorem (every element of the subalgebra generated by g is a value of a Rieger–Nishimura term in g). (4) *Converse failure*: uniqueness of the ordinary element does not imply one-generation; the minimal counterexample has eight elements, is unique at that cardinality, and is isomorphic to Z_6 with a three-element chain stacked above it. The construction class was independently identified by Citkin. (5) *Arithmetic instance*: the subgroup lattice of $\mathbb{Z}/12\mathbb{Z}$ (equivalently the divisor lattice of 12) is one-generated as a Heyting algebra — by the two-element subgroup $\{0, 6\}$ — and is therefore isomorphic to Z_6 . For every $n = p_1^{a_1} \cdots p_k^{a_k}$, the number of ordinary elements of the subgroup lattice of $\mathbb{Z}/n\mathbb{Z}$ is exactly $\prod (a_i + 1) - \prod a_i - 2^k + 1$; consequently an ordinary element exists iff n has at least two prime divisors and is not squarefree, and is *unique* iff $n = p^2 q$ — with no restriction on the number of prime factors. The least such n is 12. (6) *Substitution monoid*: endomorphisms of a one-generated Heyting algebra are determined by their value at the generator, and the identity of the endomorphism monoid is the unique endomorphism fixing the unique ordinary element; $\text{End}(Z_6) \cong Z_6$ with exactly six endomorphisms, each satisfying $f^3 = f$ (a machine-checked instance of Ruitenburg’s periodicity). In the pitch-class reading of $\mathbb{Z}/12\mathbb{Z}$, the unique ordinary element is the tritone, and the four regions of (1) are inhabited by the tritone, the augmented triad, the diminished seventh chord, and the whole-tone scale. All principal results are kernel-checked in Lean 4 against Mathlib4 with standard axioms only and no `sorry`.

1. INTRODUCTION

The free Heyting algebra on one generator — the Rieger–Nishimura lattice (Rieger 1957; Nishimura 1960) — is among the most completely understood objects in intuitionistic logic. Citkin (2024) gives a modern algebraic treatment, showing in particular that for each $n > 1$ there is exactly one

one-generated Heyting algebra Z_n of cardinality n , arising as the finite quotients (truncations) of the free algebra. Citkin’s paper isolates the class of elements he calls *ordinary* — neither regular nor dense — and observes that Z_2 through Z_5 contain none, while the generator of Z_n for $n \geq 6$ is ordinary.

This note collects a group of results that grew from a simple question: at which elements a of a Heyting algebra is the four-region lattice decomposition

$$\text{below } a \cdot \text{ below } \neg a \cdot \text{ below } \neg\neg a \text{ but not below } a \cdot \text{ meeting both } a \text{ and } \neg a$$

non-degenerate, in the sense that all four regions are simultaneously inhabited? The answer (Theorem 3.2) is: exactly at the ordinary elements. Ordinarity, introduced by Citkin as a structural condition on generators, is thus equivalently the non-degeneracy condition for a natural partition of the algebra — the partition studied in a companion preprint in a topos-theoretic setting (Brink 2026); the present note is purely lattice-theoretic and self-contained.

From the non-degeneracy theorem, several consequences follow that we have not found in the literature, and one that we subsequently found *stated* but not proved:

- **Theorem 4.1 (threshold).** Any Heyting algebra with an ordinary element has at least six elements. This generalizes Citkin’s Proposition 4(c) (2024), which bounds one-generated algebras with ordinary generators, to arbitrary Heyting algebras and arbitrary ordinary elements, with an abstract proof.
- **Theorem 4.2 (order-embedding).** For any ordinary g in any Heyting algebra H , the map $\perp \mapsto \perp$, $g \mapsto g$, $\neg g \mapsto \neg g$, $\neg\neg g \mapsto \neg\neg g$, $g \vee \neg g \mapsto g \vee \neg g$, $\top \mapsto \top$ is an order-embedding of Z_6 into H . The embedding is *not* in general a Heyting-subalgebra embedding (Remark 5.4 gives the counterexample in the free algebra).
- **Theorem 5.2 (uniqueness).** In any Heyting algebra one-generated by an ordinary element g , the element g is the *unique* ordinary element. The statement appears, without proof, in Citkin (2025, p. 13); Citkin has confirmed (personal communication, 2026) that he knows of no written proof. The machine-checked proof given here proceeds through a machine-checked proof of Nishimura’s one-variable normal-form theorem (Theorem 5.1), which may be of independent interest as a formalization.
- **Remark 5.5 (the substitution monoid).** Endomorphisms of a one-generated Heyting algebra are determined by their value at the generator, and — when the generator is ordinary — the identity of the endomorphism monoid is the unique endomorphism fixing the unique ordinary element. At Z_6 the correspondence is onto: $\text{End}(Z_6) \cong Z_6$, six endomorphisms, composition = substitution, every f satisfying $f^3 = f$.

- **Theorem 6.1 (converse failure).** Unique-ordinary does not imply one-generated: there is an eight-element Heyting algebra H_8 with a unique ordinary element that is not one-generated, and eight is the minimal such cardinality. H_8 is Z_6 with a three-element chain stacked above its top; the construction class was independently identified by Citkin (personal communication, 2026) from the statement alone.
- **Theorem 7.1 (the arithmetic instance).** The subgroup lattice of $\mathbb{Z}/12\mathbb{Z}$ is one-generated as a Heyting algebra by its two-element subgroup, hence is isomorphic to Z_6 by Citkin’s uniqueness theorem; its unique ordinary element is that subgroup. In the pitch-class reading this subgroup is the tritone $\{0, 6\}$, and the four regions of the non-degeneracy theorem are inhabited by four familiar objects of music theory (Section 9).
- **Theorem 8.1 (the p^2q law).** For n with at most two prime divisors, the subgroup lattice of $\mathbb{Z}/n\mathbb{Z}$ contains an ordinary element iff n is divisible by p^2 for some prime p with a second prime present, and contains a unique ordinary element iff $n = p^2q$ (p, q distinct primes). The least n of this form is 12. Prime-power and squarefree lattices never contain ordinary elements — the former abstractly for all chains, covering all p^k at once.
- **Theorem 8.2 (the general k -prime law and the count formula).** For arbitrary $n = p_1^{a_1} \cdots p_k^{a_k}$, the number of ordinary elements of the divisor lattice is exactly $\prod (a_i + 1) - \prod a_i - 2^k + 1$, and uniqueness forces $n = p^2q$ outright: three or more prime factors always produce at least two ordinary elements. The restriction to two primes in Theorem 8.1 is thus removed abstractly, not by extension of instances.

The results are individually modest; their interest is in the convergence. The smallest one-generated Heyting algebra with an ordinary element (Z_6 , forced by logic) and the smallest cyclic-group subgroup lattice with a unique ordinary element ($n = 12$, forced by arithmetic) are the *same six-element algebra with the same distinguished element* — and that algebra is the pitch-class-set universe of twelve-tone equal temperament, with the tritone as the distinguished element. Neither selection involves a choice: the algebra is forced from two independent directions.

Formalization. All principal results are formalized in Lean 4 against Mathlib4 and kernel-checked; axiom audits report at most `propext`, `Classical.choice`, `Quot.sound` (several results need only `propext`), and no proof contains `sorry`. Section 10 gives the theorem-by-theorem correspondence. The source is public (<https://github.com/thefalsework/papers>, directory `lean/FalseWorkPapers/Examples/`).

Prior-art standing. The prior-art situation was adjudicated by correspondence with Citkin in June 2026 (cited with permission as: A. Citkin, personal communication, 2026-06-11) and is stated precisely where each result appears. In summary: the threshold facts for one-generated algebras are Citkin’s (2024); the uniqueness statement is Citkin’s (2025, asserted without proof) and the written proof is apparently new; the converse failure is new as a statement but was reconstructed by Citkin in a few lines once posed, so we regard it as an observation, not an advance; the $\mathbb{Z}/12\mathbb{Z}$ identification and the p^2q characterization did not surface any prior art through the correspondence, though transmission difficulties (Section 11) mean their novelty is unadjudicated rather than established. Kurenniemi (2004) independently used divisor lattices as models of musical structure, in just intonation and without the Heyting layer; Section 9 records the relation.

2. PRELIMINARIES

2.1. Heyting algebras; regular, dense, and ordinary elements. A *Heyting algebra* H is a bounded distributive lattice with a binary operation $\rightarrow$ (relative pseudocomplement) satisfying $c \leq a \rightarrow b$ iff $c \wedge a \leq b$. The *pseudocomplement* is $\neg a = a \rightarrow \perp$. Standard identities used throughout: $a \wedge \neg a = \perp$; $a \leq \neg \neg a$; $\neg a = \neg \neg \neg a$; $\neg \neg$ is a monotone idempotent operator.

Following Citkin (2024, §2.1): an element $a \in H$ is *regular* if $\neg \neg a = a$, *dense* if $\neg a = \perp$, and **ordinary** if it is neither. Equivalently, a is ordinary iff

$$a \neq \perp \quad \text{and} \quad \neg a \neq \perp \quad \text{and} \quad \neg \neg a \neq a.$$

($a = \perp$ is regular, so the first clause is implied by the third and second in context; we state all three because that is the form the non-degeneracy criterion uses. The three clauses correspond to the classification of elements as bottom-like, dense, or regular: an element is ordinary precisely when it escapes all three.) In a Boolean algebra every element is regular, so Boolean algebras have no ordinary elements.

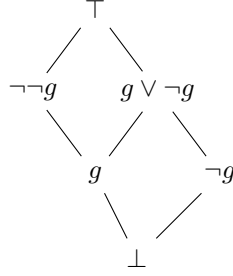
2.2. The Rieger–Nishimura lattice and the algebras Z_n . The free Heyting algebra $F(1)$ on one generator p is the Rieger–Nishimura lattice: its elements are $\perp, \top$, and the values of the *Nishimura terms* — the ascending ladder generated from p by the recursion (in one standard indexing)

$$g_0 = p, \quad g_1 = \neg p, \quad f_1 = p \vee \neg p, \quad g_2 = \neg \neg p, \quad g_3 = \neg \neg p \rightarrow p,$$

$$g_{n+4} = g_{n+3} \rightarrow (g_n \vee g_{n+1}), \quad f_{n+2} = g_{n+2} \vee g_{n+1}.$$

By Citkin (2024, Theorem 6), for each $n > 1$ there is exactly one one-generated Heyting algebra of cardinality n , denoted Z_n ; the Z_n are the finite quotients of $F(1)$. The first non-trivial ones:

- $Z_2 = \{\perp, \top\}$ (Boolean), Z_3 = the three-element chain, Z_4, Z_5 — none of which contain an ordinary element (Citkin 2024);
- Z_6 , with elements $\perp, g, \neg g, \neg \neg g, g \vee \neg g, \top$ and Hasse diagram



(the two incomparabilities are $\neg g \parallel \neg\neg g$ and $\neg\neg g \parallel g \vee \neg g$; note $g \leq \neg\neg g$ and both $g, \neg g \leq g \vee \neg g$).

We write “ H is one-generated by g ” for: the smallest subset of H containing g and closed under $\wedge, \vee, \rightarrow, \neg, \perp, \top$ is H itself.

3. THE FOUR-REGION DECOMPOSITION AND THE NON-DEGENERACY THEOREM

Definition 3.1. Fix a Heyting algebra H and an element $a \in H$. For $x \in H$ with $x \neq \perp$, say x lies in region

- I** if $x \leq a$;
- R** if $x \leq \neg a$;
- E** if $x \leq \neg\neg a$ and $x \not\leq a$;
- D** if $x \wedge a \neq \perp$ and $x \wedge \neg a \neq \perp$.

Proposition 3.1 (partition). *For every $x \in H$ with $x \neq \perp$, exactly one of I, R, E, D holds. (Lean: `lattice_four_position_partition`.)*

Proof. Disjointness. I–R: $x \leq a \wedge \neg a = \perp$. I–E: E requires $x \not\leq a$. I–D: $x \leq a$ gives $x \wedge \neg a \leq a \wedge \neg a = \perp$. R–E and E–D: $x \leq \neg\neg a$ gives $x \wedge \neg a \leq \neg\neg a \wedge \neg a = \perp$, which contradicts $x \leq \neg a$ with $x \neq \perp$ (for R) and the second conjunct of D. R–D: $x \leq \neg a$ gives $x \wedge a = \perp$. *Exhaustiveness.* Case on whether $x \wedge a$ and $x \wedge \neg a$ are $\perp$. If both are non- $\perp$, $x \in D$. If $x \wedge \neg a = \perp$ then $x \leq \neg\neg a$, and $x \in I$ or $x \in E$ according as $x \leq a$ or not. If $x \wedge a = \perp$ then $x \leq \neg a$, so $x \in R$. If both are $\perp$ then $x \leq \neg a \wedge \neg\neg a = \perp$, contradicting $x \neq \perp$. $\square$

The decomposition is trivial for many choices of a : if $a = \perp$, regions I, E, D are empty; if a is dense, R is empty; if a is regular, E is empty ($x \leq \neg\neg a = a$ puts x in I). The exact non-degeneracy condition is:

Theorem 3.2 (non-degeneracy). *In any Heyting algebra H and for any $a \in H$, all four regions are simultaneously inhabited if and only if a is ordinary:*

$$a \neq \perp \wedge \neg a \neq \perp \wedge \neg\neg a \neq a.$$

(Lean: `allFourCellsInhabited_iff`.)

Proof. ($\Leftarrow$) The four witnesses are $a \in I$, $\neg a \in R$, $\neg\neg a \in E$ (it is $\leq \neg\neg a$; if $\neg\neg a \leq a$ then $\neg\neg a = a$, contradiction), and $a \vee \neg a \in D$ (its meets with a

and $\neg a$ are $a \neq \perp$ and $\neg a \neq \perp$). ($\Rightarrow$) An inhabitant of I forces $a \neq \perp$; of R forces $\neg a \neq \perp$; of E forces $\neg\neg a \neq a$ (an $x \leq \neg\neg a$ with $x \not\leq a$ is impossible when $\neg\neg a = a$). $\square$

Two immediate corollaries situate the condition:

Corollary 3.3 (Boolean collapse). *No Boolean algebra contains an ordinary element; hence the decomposition is never non-degenerate in a Boolean algebra. Moreover the double-negation map $a \mapsto \neg\neg a$ of any Heyting algebra lands only on non-ordinary (regular) elements, so the Boolean algebra of regular elements — the classical shadow of H under Glivenko’s translation — never carries a non-degenerate decomposition either.* (Lean: `boolean_no_kernel`, `glivenko_collapse`.)

Corollary 3.4 (dense-bottom lemma). *If H has a least non-zero element (an atom below every non-zero element), then H has no ordinary elements: every non-zero element is dense.* (Lean: `no_ordinary_of_least_nonzero`.)

4. THE SIX-ELEMENT THRESHOLD AND THE ORDER-EMBEDDING

Theorem 4.1 (threshold). *Let H be a Heyting algebra containing an ordinary element g . Then the six elements*

$$\perp, \quad g, \quad \neg g, \quad \neg\neg g, \quad g \vee \neg g, \quad \top$$

are pairwise distinct; hence $|H| \geq 6$ (in particular any finite Heyting algebra with a non-degenerate four-region decomposition has at least six elements). (Lean: `ordinary_card_ge_six`, `allFourCells_card_ge_six`.)

Proof sketch. Most distinctions are immediate from ordinariness ($g \neq \perp$, $\neg g \neq \perp$, $\neg\neg g \neq g$) and the basic identities ($g \wedge \neg g = \perp$ separates g from $\neg g$ since both are non- $\perp$; $\neg\neg g \wedge \neg g = \perp$ similarly). The load-bearing case is $g \vee \neg g \neq \top$: if $g \vee \neg g = \top$ then, by distributivity, $\neg\neg g = \neg\neg g \wedge (g \vee \neg g) = (\neg\neg g \wedge g) \vee (\neg\neg g \wedge \neg g) = g \vee \perp = g$, contradicting ordinariness. $\square$

Relation to prior art. For *one-generated* algebras with ordinary *generators* this is Citkin’s Proposition 4(c) (2024), and the non-existence of ordinary elements in Z_2 – Z_5 is observed in the same paper. Theorem 4.1 removes both restrictions: any ordinary element, any Heyting algebra, abstract proof with no enumeration. The Z_5 case then follows as a corollary rather than a check (the formalization derives it both ways and confirms agreement).

Theorem 4.2 (order-embedding). *With H and g as above, the assignment*

$$\perp \mapsto \perp, \quad g \mapsto g, \quad \neg g \mapsto \neg g, \quad \neg\neg g \mapsto \neg\neg g, \quad (g \vee \neg g) \mapsto g \vee \neg g, \quad \top \mapsto \top$$

*defines an order-embedding $Z_6 \hookrightarrow H$: the six images realize exactly the Hasse structure of Z_6 , including the two incomparabilities $\neg g \parallel \neg\neg g$ and $\neg\neg g \parallel g \vee \neg g$ (each of the 18 non-relations is forced by ordinariness). The axiom audit for this theorem reports **propext** only.* (Lean: `div12OrderEmbedding`.)

Scope guard. Theorem 4.2 is an *order*-embedding, not a Heyting-subalgebra embedding: the six-element image is in general not closed under $\rightarrow$. Remark 5.4 exhibits the failure in $F(1)$ itself. (A note on the Lean name: `div12OrderEmbedding` is stated for an arbitrary Heyting algebra H and arbitrary ordinary g — the name refers to the *source* of the embedding, $\text{Div12} \cong Z_6$, not to an instance restriction.) The subalgebra-level statement that *is* true is Citkin’s Proposition 3.1 (2025): any Heyting algebra with an ordinary element contains a subalgebra isomorphic to some Z_n , $n \geq 6$ — the subalgebra generated by the ordinary element, which by Theorem 5.1 below is a quotient of the Rieger–Nishimura ladder, not necessarily the six-element truncation.

5. NISHIMURA NORMAL FORM AND THE UNIQUENESS OF THE ORDINARY ELEMENT

The engine of the uniqueness proof is the classical one-variable normal-form theorem, which we formalize in full:

Theorem 5.1 (Nishimura normal form, machine-checked). *Let H be a Heyting algebra and $g \in H$. Every element of the Heyting subalgebra generated by g is $\top$ or a value at g of a Nishimura term.* (Lean: `generatedBy_isLadderValue`.)

Proof structure (as formalized). The set of ladder values is shown closed under each Heyting operation in turn: the complement table ($\neg x_n$ is again a ladder value, with $\neg x_n = \perp$ from index 4 up); the join and meet tables (comparable pairs are absorbed; the incomparable diagonals are the recursion identities $x_m \vee x_{m+1} = x_{m+3}$ and $x_m \vee x_{m+2} = x_{m+5}$ for m odd, dually for meets); and the implication table, whose intricacy is the real content. The decisive simplification is that implications into *odd-indexed* targets are terminal — $x_m \rightarrow x_{2i+3} = x_{2i+3}$ whenever $x_{2i+1} \leq x_m$, by currying — so the only genuinely recursive family is implication into even targets, which climbs by two with a comparability ceiling; a lexicographic measure closes the induction, and the Lean kernel certifies termination. Structural induction over the generated subalgebra then gives the theorem. Axioms: `propext`, `Quot.sound`; no `Classical.choice`. $\square$

The theorem itself is classical (Nishimura 1960; algebraic proof in Citkin 2024). Its formal value here is that it converts “one-generated” into an induction principle, discharging what was previously a citation from the following result:

Theorem 5.2 (uniqueness of the ordinary element). *Let H be a Heyting algebra one-generated by g , and suppose g is ordinary. Then g is the unique ordinary element of H . Consequently every Z_n with $n \geq 6$, and the free Heyting algebra $F(1)$ itself, has its generator as unique ordinary element — equivalently (Theorem 3.2), as the unique element at which the*

four-region decomposition is non-degenerate. (Lean: `nishimura_ordinary_unique`, `nishimura_kernel_unique_of_generated`.)

Proof idea. By Theorem 5.1 every element is a ladder value; the ladder is then analyzed directly. Every rung from index 4 up lies above $g \vee \neg g$ and is therefore dense (its pseudocomplement is $\perp$); the rungs $\neg g$ and $\neg\neg g$ are regular; $\perp$ and $\top$ are regular and dense respectively. Only the generator’s rung survives. Verified additionally by unconditional exhaustive computation at $n = 6, 7, 8$. Axioms: `propext`, `Quot.sound`. $\square$

Prior-art standing (adjudicated). The statement of Theorem 5.2 appears in print in Citkin (2025, p. 13): “the algebra Z_n contains a single ordinary element that generates the entire algebra” — asserted without proof, feeding his Proposition 3.1. In correspondence, Citkin stated he is “not aware of any known proofs” and sketched the folklore argument specialists would give: the free algebra $F(1)$ has a single ordinary element; the finite Z_n are its quotients; regularity and density are preserved under Heyting quotients; hence uniqueness passes down. That argument presupposes the single-ordinary-element property of $F(1)$ itself, which is exactly what the ladder analysis above establishes; the two routes are consistent. The standing of Theorem 5.2 is therefore: *statement published by Citkin as an unproved observation; the machine-checked proof presented here apparently the first written proof.*

Corollary 5.3 (unique generator). *A one-generated Heyting algebra of cardinality ≥ 6 has a unique generator. (Generators are ordinary by Citkin’s Proposition 4(c); the ordinary element is unique by Theorem 5.2. The general statement was communicated by Citkin; the instance at Z_6 is kernel-checked directly as `Div12.generator_unique`, by trapping each non-generator inside an explicit proper subalgebra.)*

Remark 5.4 (the subalgebra upgrade of Theorem 4.2 is false in general). In $F(1)$, with g the free generator, $\neg\neg g \rightarrow g$ is the fifth rung of the ladder — a genuinely new element outside the six-element set $\{\perp, g, \neg g, \neg\neg g, g \vee \neg g, \top\}$. So the order-embedding of Theorem 4.2 does not preserve $\rightarrow$ in general; it is a Heyting embedding exactly when the ladder above index 4 collapses into the six-set (equivalently, when the subalgebra generated by g has six elements). This is a kernel-checked counterexample, recorded to prevent the natural overreading of Theorem 4.2.

Remark 5.5 (the substitution monoid). The Heyting endomorphisms of H form a monoid $\text{End}(H)$ under composition, and the normal-form machinery of Theorem 5.1 controls it. Homs commute with the term ladder — $f(x_n(g)) = x_n(fg)$, index by index — so two homs out of a one-generated algebra that agree on the generator agree everywhere: evaluation at g embeds $\text{End}(H)$ into H . If moreover g is ordinary, the monoid identity acquires an order-theoretic characterization: *an endomorphism is the identity iff it fixes the unique ordinary element* (by Theorem 5.2 there is exactly one). At Z_6

the evaluation embedding is onto: every endomorphism is the substitution $g \mapsto b$ for a unique b , so $\text{End}(\mathbf{Z}_6) \cong \mathbf{Z}_6$ — exactly six endomorphisms, with composition transported to the substitution product $(b, c) \mapsto c[g := b]$. Three further facts are kernel-checked there: no non-identity endomorphism preserves ordinariness (every proper substitution lands the generator in a degenerate region); every endomorphism satisfies $\mathbf{f}^3 = \mathbf{f}$ — a machine-checked one-variable instance of Ruitenburg’s periodicity theorem (Ruitenburg 1984), with the substitution at $\neg g$ the unique element of period exactly two (it swaps with the one at $\neg\neg g$); and the group of units is trivial. (Lean: `heytingHom_map_nishimuraTerm`, `heytingHom_ext_of_generated`, `endo_eq_id_iff_fixes_kernel`, `Div12.endo_bijection`, `Div12.subst_cube`; all in the file `SubstitutionMonoid.lean`, axioms `propext`, `Quot.sound`.)

6. THE CONVERSE FAILS: H_8

Theorem 5.2 says one-generated (with ordinary generator) $\Rightarrow$ unique ordinary element. The converse is false, and the minimal counterexample is small:

Theorem 6.1 (converse failure). *There is an eight-element Heyting algebra H_8 with a unique ordinary element a such that H_8 is not one-generated by a (nor by any element). Eight is minimal: every Heyting algebra of cardinality ≤ 7 with a unique ordinary element is one-generated — concretely, is Z_6 or Z_7 , since Z_2 – Z_5 contain no ordinary element. H_8 is unique at cardinality eight up to isomorphism.* (Lean: `unique_ordinary_converse_false`.)

Construction. H_8 is Z_6 with a three-element chain stacked above its top (the top of Z_6 identified with the bottom of the chain): $6 + 3 - 1 = 8$ elements. The two new elements above the old top are dense, so no new ordinary elements are created and the unique ordinary element of Z_6 survives; but the term ladder of a stabilizes after seven values and never reaches the eighth element, so a does not generate (machine-checked: the term values of a stay inside a proper subalgebra — which, per Citkin’s Proposition 3.1, is precisely a Z_7 core; the isomorphism of the core with Z_7 is itself kernel-checked, `h8_ladder_core`). $\square$

Status of the minimality claim. The non-generation and the Z_7 -core anatomy are kernel-checked. The *minimality* (no counterexample below eight elements, uniqueness at eight) rests on an exhaustive enumeration of finite Heyting algebras of cardinality ≤ 12 — via Birkhoff duality, as downset lattices of finite posets, with per-size counts cross-checked against OEIS A006982 — performed by script, not formalized in Lean. We flag the different evidential grade explicitly.

Independent confirmation. Posed the bare statement, Citkin reconstructed the construction class from first principles without seeing the witness: glue a one-generated six-element algebra to a chain of height ≥ 3 ; new chain elements are dense; the unique ordinary element survives; the

result is not one-generated; smallest case $Z_6 + Z_3$, eight elements (personal communication, 2026-06-11). His reconstruction matches H_8 exactly. He regarded the fact as readily derivable once posed; we therefore present Theorem 6.1 as an observation with a machine-checked witness, not as a field advance.

Proposition 6.2 (dense-filter salvage). *In any Heyting algebra with a unique ordinary element a , the dense elements are exactly the filter above $a \vee \neg a$. (Lean: `unique_ordinary_dense_iff`.)*

Proof sketch (following the formalization). Let x be dense. Meeting with a dense element does not change complements: $\neg(b \wedge x) = \neg b$. Hence $\neg(a \wedge x) = \neg a \neq \perp$ and $\neg\neg(a \wedge x) = \neg\neg a$, which cannot equal $a \wedge x$ (that would force $\neg\neg a \leq a$, collapsing a 's ordinariness); so $a \wedge x$ is ordinary, hence $a \wedge x = a$ by uniqueness, i.e. $a \leq x$. For $\neg a$: the element $\neg a \wedge x$ has complement $\neg\neg a \neq \perp$ (since $a \neq \perp$), so if it were not regular it would be ordinary, hence equal to a by uniqueness — forcing $a \leq \neg a$ and so $a = \perp$, a contradiction. So $\neg a \wedge x$ is regular: $\neg a \wedge x = \neg\neg(\neg a \wedge x) = \neg\neg\neg a = \neg a$, i.e. $\neg a \leq x$. Hence $a \vee \neg a \leq x$. Conversely anything above $a \vee \neg a$ is dense, since $\neg(a \vee \neg a) = \neg a \wedge \neg\neg a = \perp$. $\square$

The statement was independently confirmed by Citkin (personal communication, 2026), by an argument of the same shape. The least dense element is thus $a \vee \neg a$ — the fourth Nishimura term.

The pair (Z_8, H_8) is instructive: same cardinality, both with a unique ordinary element, only one of them one-generated. Uniqueness of the ordinary element does not certify that an algebra is a truncation of the free algebra; the two properties of Z_6 established in the next section — unique ordinary element *and* one-generation — are independent facts requiring separate proof.

7. THE SUBGROUP LATTICE OF $\mathbb{Z}/12\mathbb{Z}$ IS Z_6

Let $\text{Div}12$ denote the divisor lattice of 12 — elements $\{1, 2, 3, 4, 6, 12\}$, order = divisibility, meet = gcd, join = lcm — which is isomorphic to the subgroup lattice of the cyclic group $\mathbb{Z}/12\mathbb{Z}$ under $d \mapsto$ (the unique subgroup of order d). $\text{Div}12$ is a finite distributive lattice, hence a Heyting algebra.

Theorem 7.1 (the arithmetic instance). *$\text{Div}12$ is one-generated as a Heyting algebra by the divisor 2 (equivalently, the two-element subgroup $\{0, 6\} \leq \mathbb{Z}/12\mathbb{Z}$). Explicitly, with $g = 2$:*

$$\neg g = 3, \quad \neg\neg g = 4, \quad g \vee \neg g = 6, \quad g \wedge \neg g = 1 = \perp, \quad \neg\neg g \vee (g \vee \neg g) = 12 = \top.$$

(Lean: `Div12.one_generated_by_tritone`, `Div12.tritone_generates`.)

Hence, by Citkin's uniqueness theorem (2024, Theorem 6), $\mathbf{Div}12 \cong \mathbf{Z}_6$, with the divisor 2 as the (unique, by Corollary 5.3) free generator. By Theorem 5.2 the divisor 2 is also the unique ordinary element of $\text{Div}12$; the formalization derives this three independent ways (exhaustive check,

the uniqueness law, and through the generation machinery), and the three agree.

The proof is a finite computation, but the statement has content: it identifies a lattice arising in nature (subgroups of a cyclic group; pitch-class sets closed under transposition, in the reading of Section 9) with the smallest non-degenerate truncation of the free Heyting algebra on one generator, and its distinguished subgroup with the free generator. Neither side of the identification was constructed to meet the other.

8. DIVISOR LATTICES OF $\mathbb{Z}/n\mathbb{Z}$: THE p^2q CHARACTERIZATION

Which cyclic groups have subgroup lattices containing ordinary elements? The subgroup lattice of $\mathbb{Z}/n\mathbb{Z}$ is the divisor lattice of n , which is a product of chains — one chain per prime divisor, of length (exponent + 1). (This structure theorem is standard and is cited, not re-proved, for general n ; the instance at $n = 12$, $\text{Div}12 \cong C_3 \times C_2$ as Heyting algebras, is kernel-checked with $\rightarrow$ and $\neg$ preserved, `div12OrderIsoChains`.)

Theorem 8.1 (the p^2q law).

- (1) *A totally ordered Heyting algebra contains no ordinary element (every non- $\perp$ element has pseudocomplement $\perp$, i.e. is dense). Hence no prime-power $n = p^k$ yields an ordinary element — abstractly, for all k .*
- (2) *In a product of two non-trivial bounded chains, the ordinary elements are exactly the pairs with one coordinate $\perp$ and the other strictly between $\perp$ and $\top$ of its chain.*
- (3) *For $n = p^a q^b$ ($a, b \geq 1$, $p \neq q$ prime): the divisor lattice of n contains an ordinary element iff $\max(a, b) \geq 2$. In particular squarefree $n = pq$ (6, 10, 15, ...) yield none.*
- (4) *The ordinary element is unique iff $\{a, b\} = \{2, 1\}$, i.e. iff $n = p^2q$. The least such n is 12.*
- (5) *Boundary behavior, kernel-checked at instances: $n = 24 = 2^3 \cdot 3$ has two ordinary elements, $n = 36 = 2^2 \cdot 3^2$ has two, $n = 60 = 2^2 \cdot 3 \cdot 5$ has three.*

(Lean: `total_no_kernel`, `chainProd_kernel_exists_iff`, `chainProd_kernel_unique_iff`, `why_twelve`.)

Under the explicit isomorphism $\text{Div}12 \cong C_3 \times C_2$, the unique ordinary element of the $n = 12$ lattice is carried to $(1, 0)$ — and back in $\text{Div}12$ it is the divisor 2, the generator of Theorem 7.1.

Theorem 8.2 (the general k -prime law and the count formula). *Let $n = p_1^{a_1} \cdots p_k^{a_k}$ ($k \geq 1$, all $a_i \geq 1$, p_i distinct primes), with divisor lattice the product of chains $C_{a_1+1} \times \cdots \times C_{a_k+1}$ under the pointwise Heyting structure. Then, abstractly in k and the exponent vector:*

- (1) *The ordinary elements are exactly the points with some coordinate $\perp$ and some coordinate strictly internal to its chain (generalizing Theorem 8.1(2)).*
- (2) *An ordinary element exists iff $k \geq 2$ and some $a_i \geq 2$.*
- (3) *The number of ordinary elements is exactly $\prod_i (a_i + 1) - \prod_i a_i - 2^k + 1$ (inclusion–exclusion on the complement of (1): the points with no $\perp$ coordinate, $\prod_i a_i$ of them, and the points with every coordinate in $\{\perp, \top\}$, 2^k of them, overlap exactly at $\top$).*
- (4) *The ordinary element is unique iff $k = 2$ and $\{a_1, a_2\} = \{2, 1\}$ — iff $n = p^2q$ — now with no restriction on the number of prime factors: with $k \geq 3$, an internal value at one coordinate can pair with $\perp$ placed at either of two other coordinates (with or without a $\top$ elsewhere), so a kernel is never alone.*

(Lean: `piChain_kernel_iff`, `piChain_kernel_exists_iff`, `piChain_kernel_unique_iff`, `piChain_kernel_card`, in the file `WhyTwelveGeneral.lean`.)

Two-prime instance of (3): $(a+1)(b+1) - ab - 4 + 1 = (a-1) + (b-1)$, recovering Theorem 8.1(3)–(5): existence iff the count is positive iff $\max(a, b) \geq 2$; uniqueness iff the count is 1 iff $\{a, b\} = \{2, 1\}$; $24 = 2^3 \cdot 3 \rightarrow 2$ and $36 = 2^2 \cdot 3^2 \rightarrow 2$. At $60 = 2^2 \cdot 3 \cdot 5$ the formula gives $3 \cdot 2 \cdot 2 - 2 - 8 + 1 = 3$, matching the exhaustive check (`sixty_kernel_three`); squarefree n with any number of primes gives $2^k - 1 - 2^k + 1 = 0$. Note the naive coordinate-wise sum $\sum_i (a_i - 1)$ fails for $k \geq 3$ (it gives 1 at 60): an ordinary element needs only *some* coordinate at $\perp$ alongside a strictly internal one, not all remaining coordinates at $\perp$. The count instances at 12, 24, 36, 60, and $210 = 2 \cdot 3 \cdot 5 \cdot 7$ are additionally verified by exhaustive computation.

Scope statements, fixed precisely. (i) Theorem 8.2 is proved abstractly for all k and all exponent vectors; what remains cited rather than proved is the classical structure theorem “the divisor lattice of n is the product of one chain per prime” for general n (the instance at $n = 12$ is kernel-checked, `div12OrderIsoChains`). (ii) $18 = 2 \cdot 3^2$ and $20 = 2^2 \cdot 5$ are also of the form p^2q and also carry unique ordinary elements — necessarily the *same* six-element lattice. Twelve is the **least** n of the form, not the only one. (iii) The identification of n -tone equal temperament with $\mathbb{Z}/n\mathbb{Z}$ (next section) is a modeling convention, not mathematics.

The two-sided forcing. Theorems 5.2 and 8.1–8.2 select the same object from independent directions. On the logic side, Z_6 is the *first* one-generated Heyting algebra whose four-region decomposition is non-degenerate (Theorem 4.1: nothing below six elements qualifies), with the free generator as the forced distinguished element. On the arithmetic side, $n = 12$ is the *first* cyclic-group order whose subgroup lattice carries an ordinary element at all, and — 12 being of the uniqueness shape p^2q — carries exactly one. The two firsts coincide: $Z_6 \cong \text{Div}12 \cong C_3 \times C_2$, free generator = divisor 2 = $(1, 0)$.

9. A REMARK ON THE MUSICAL READING

This section is interpretive and is flagged as such; nothing in Sections 3–8 depends on it.

Identify $\mathbb{Z}/12\mathbb{Z}$ with the pitch classes of twelve-tone equal temperament. The subgroups are then the transpositionally symmetric pitch-class sets: $\{0\}$ (trivial), $\{0, 6\}$ (the **tritone** dyad), $\{0, 4, 8\}$ (the **augmented triad**), $\{0, 3, 6, 9\}$ (the **diminished seventh** chord), $\{0, 2, 4, 6, 8, 10\}$ (the **whole-tone** scale), and $\mathbb{Z}/12\mathbb{Z}$ (the chromatic aggregate). Under Theorem 7.1 the unique ordinary element — the free generator of the lattice qua Z_6 — is the tritone. The four regions of the non-degeneracy theorem (Theorem 3.2) at the tritone are inhabited by:

region	witness	pitch-class object
I (below g)	g	tritone $\{0, 6\}$
R (below $\neg g$)	$\neg g$	augmented triad $\{0, 4, 8\}$
E (below $\neg\neg g$, not below g)	$\neg\neg g$	diminished seventh $\{0, 3, 6, 9\}$
D (meets both)	$g \vee \neg g$	whole-tone scale $\{0, 2, 4, 6, 8, 10\}$

That the diminished seventh — the pivot chord of tonal ambiguity in common-practice harmony — occupies the double-negation residue $\neg\neg g \setminus g$, the region that exists only in non-Boolean logic, is at minimum a striking coincidence of structure; we do not claim more for it here. A companion series of framework papers (Brink 2026, and references therein) develops an interpretive program on top of this instance; the present note is confined to the mathematics.

Prior art in the musical direction. Kurenniemi (2004) independently proposed divisor lattices as models of musical structure, working in 5-limit just intonation with the divisor lattices of large numbers (e.g. 8640) under gcd/lcm, and observing the near-coincidence $2^{19} \approx 3^{12}$ (the Pythagorean comma). His treatment stops at the distributive lattice and its geometric embedding: no Heyting structure, no relative pseudocomplement, no one-generation, no ordinary elements. The results of Sections 7–8 do not appear there; we cite it as independent precedent for the divisor-lattice formalism in music, with the observation that his machinery aimed at $n = 12$ rather than just-intonation integers would meet the present construction.

10. FORMALIZATION

All principal results are formalized in Lean 4 against Mathlib4 (toolchain v4.30.0-rc2 era) and kernel-checked. No proof contains `sorry`; no proof uses `native_decide`. Axiom audits (`#print axioms`) report:

Result	Lean theorem	File (Examples/)	Axioms
Partition (Prop. 3.1)	<code>lattice_four_position_partition</code>	<code>Lattice/</code> <code>FourPositionLattice.</code> <code>lean[†]</code>	all three
Non-degeneracy (Thm 3.2)	<code>allFourCellsInhabited_</code> <code>iff</code>	<code>NishimuraKernelLaw.</code> <code>lean</code>	<code>propext</code>
Boolean/Glivenko (Cor. 3.3)	<code>boolean_no_kernel,</code> <code>glivenko_collapse</code>	<code>GlivenkoCollapse.lean</code>	<code>propext</code>
Dense-bottom (Cor. 3.4)	<code>no_ordinary_of_least_</code> <code>nonzero</code>	<code>LadderCore.lean</code>	all three
Threshold (Thm 4.1)	<code>ordinary_card_ge_six,</code> <code>allFourCells_card_ge_</code> <code>six</code>	<code>LadderCore.lean</code>	all three
Order- embedding (Thm 4.2)	<code>div12OrderEmbedding</code>	<code>LadderCore.lean</code>	<code>propext</code>
Normal form (Thm 5.1)	<code>generatedBy_</code> <code>isLadderValue</code>	<code>NishimuraNormalForm.</code> <code>lean</code>	<code>propext,</code> <code>Quot.sound</code>
Uniqueness (Thm 5.2)	<code>nishimura_ordinary_</code> <code>unique,</code> <code>nishimura_kernel_</code> <code>unique_of_generated</code>	<code>NishimuraKernelLaw.</code> <code>lean,</code> <code>NishimuraNormalForm.</code> <code>lean</code>	<code>propext,</code> <code>Quot.sound</code>
Unique generator at Z_6 (Cor. 5.3)	<code>Div12.generator_unique</code>	<code>LadderCore.lean</code>	<code>propext,</code> <code>Quot.sound</code>
Subalgebra failure (Rem. 5.4)	<code>via nishimuraTerm_himp_</code> <code>diagonal</code>	<code>NishimuraNormalForm.</code> <code>lean</code>	<code>propext,</code> <code>Quot.sound</code>
H_8 (Thm 6.1)	<code>unique_ordinary_</code> <code>converse_false,</code> <code>h8_ladder_core</code>	<code>UniqueOrdinaryConverse.</code> <code>lean, LadderCore.lean</code>	<code>propext,</code> <code>Quot.sound</code>
Dense filter (Prop. 6.2)	<code>unique_ordinary_dense_</code> <code>iff</code>	<code>UniqueOrdinaryConverse.</code> <code>lean</code>	<code>propext,</code> <code>Quot.sound</code>
$\text{Div}12 \cong Z_6$ (Thm 7.1)	<code>Div12.one_generated_by_</code> <code>tritone, Div12.tritone_</code> <code>generates,</code> <code>Div12.kernel_unique (+</code> <code>_via_law,</code> <code>_via_subalgebra)</code>	<code>NishimuraTruncations.</code> <code>lean,</code> <code>NishimuraNormalForm.</code> <code>lean</code>	<code>propext,</code> <code>Quot.sound</code>
p^2q law (Thm 8.1)	<code>total_no_kernel,</code> <code>chainProd_kernel_</code> <code>exists_iff, chainProd_</code> <code>kernel_unique_iff,</code> <code>why_twelve,</code> <code>div12OrderIsoChains</code>	<code>WhyTwelve.lean</code>	all three
k -prime law + count (Thm 8.2)	<code>piChain_kernel_iff,</code> <code>piChain_kernel_exists_</code> <code>iff, piChain_kernel_</code> <code>unique_iff,</code> <code>piChain_kernel_card</code>	<code>WhyTwelveGeneral.lean</code>	all three
Substitution monoid (Rem. 5.5)	<code>heytingHom_map_</code> <code>nishimuraTerm,</code> <code>heytingHom_ext_of_</code> <code>generated, endo_eq_id_</code> <code>iff_fixes_kernel,</code> <code>Div12.endo_bijection,</code> <code>Div12.substHom_comp,</code> <code>Div12.subst_cube</code>	<code>SubstitutionMonoid.</code> <code>lean</code>	<code>propext,</code> <code>Quot.sound</code>

“All three” abbreviates `propext`, `Classical.choice`, `Quot.sound`. [†]The partition theorem lives under `lean/FalseWorkPapers/Lattice/`, not `Examples/`.

Truncation computations at Z_5 – Z_8 are exhaustive `decide` checks over concrete decidable instances certified one-generated by exhibiting every element as a term value. One item rests partly outside the kernel and is flagged where stated: the minimality of H_8 (script-level enumeration, Section 6). The general “divisor lattice = product of chains” structure theorem and Citkin’s Theorem 6 (uniqueness of Z_n) are cited classical inputs.

Source: <https://github.com/thefalsework/papers>, in the directory `lean/FalseWorkPapers/Examples/`; audit lines are collected in `Examples/HeytingTypeInstance.lean`.

Methodological note. Several proofs were developed against the published Rieger–Nishimura structure rather than from finite models, for a reason worth recording: the finite truncations Z_n are quotients of $F(1)$, so an identity read off a finite model may be a collapse artifact — true in Z_n , false in $F(1)$ — and would compile. (Concretely: Z_8 reports $x_5 \vee x_6 = \top$, but in $F(1)$ the join is the eighth rung.) Finite models were used as sound *refuters* only, within their faithful window; every load-bearing table entry is proved abstractly.

11. PRIOR ART, CORRESPONDENCE, AND ACKNOWLEDGEMENTS

The prior-art adjudication was conducted by correspondence with Alex Citkin (June 2026, three exchanges; cited with his permission as personal communication, 2026-06-11). His 2024 paper supplies the Z_n classification, the ordinary-element notion, the Z_2 – Z_5 observation, and Proposition 4(c); his December 2025 preprint (Citkin 2025) states the uniqueness of the ordinary element without proof (p. 13) and proves Proposition 3.1 (subalgebra containment of a Z_n , $n \geq 6$, in any algebra with an ordinary element). In correspondence he confirmed knowing no written proof of the uniqueness statement, supplied the folklore quotient argument, independently reconstructed the H_8 construction class and confirmed its eight-element minimality, communicated the unique-generator fact (Corollary 5.3), and supplied an independent proof of Proposition 6.2. The identification $\text{Div}12 \cong Z_6$ and the p^2q characterization were posed to him but, owing to a terminological ambiguity in the posing (“divisor lattice of 12” read as a twelve-element algebra), were not adjudicated; their novelty standing is accordingly *no prior art surfaced, not established*. We record this rather than claim novelty.

The question that motivated the non-degeneracy theorem arose in a companion program (Brink 2026) concerning a four-fold partition of subobject lattices in `topoi`; the present note isolates everything that is pure lattice theory. Erkki Kurenniemi’s unpublished 2004 notebook was located after the fact and is cited as independent precedent for divisor-lattice models of musical structure (Section 9).

AI disclosure. This paper and its formalization were produced with AI assistance (Anthropic Claude) under the correction architecture documented in a companion methodology paper (Brink 2026b): machine verification for every principal claim, explicit evidential grading of unformalized steps, pre-registered experiments for the truncation computations, and expert correspondence for prior-art adjudication.

REFERENCES

- [1] Bezhanishvili, G., Bezhanishvili, N., & de Jongh, D. (2008). The Kuznetsov–Gerčiu and Rieger–Nishimura logics: The boundaries of the finite model property. *Logic and Logical Philosophy*, 17. (For the Nishimura polynomial recursion.)
- [2] Birkhoff, G. (1937). Rings of sets. *Duke Mathematical Journal*, 3(3), 443–454.
- [3] Brink, C. (2026). A four-position partition of morphisms in elementary topoi with distinction structure. Preprint, <https://github.com/thefalsework/papers>.
- [4] Brink, C. (2026b). Epistemic dependency as structural condition. Preprint, <https://github.com/thefalsework/papers>.
- [5] Citkin, A. (2024). An algebraic proof of the Nishimura theorem. *Logics*, 2(4), 148–157.
- [6] Citkin, A. (2025). Hereditarily structurally complete superintuitionistic logics and primitive varieties of Heyting algebras. arXiv:2512.05633.
- [7] Citkin, A. (2026). Personal communication, 2026-06-11. Cited with permission.
- [8] de Jongh, D., & Chagrova, L. (2006). ILLC Prepublication PP-2006-25, University of Amsterdam. (Nishimura polynomials, Definition 37.)
- [9] Kurenniemi, E. (2004). *Chords, scales, and divisor lattices*. Unpublished Mathematica notebook, DIMI.
- [10] Mathlib Community (2026). Mathlib4. <https://github.com/leanprover-community/mathlib4>.
- [11] Nishimura, I. (1960). On formulas of one variable in intuitionistic propositional calculus. *Journal of Symbolic Logic*, 25.
- [12] OEIS Foundation (n.d.). A006982: Number of unlabeled distributive lattices with n elements. <https://oeis.org/A006982>.
- [13] Rieger, L. (1957). A remark on the so-called free closure algebras. *Czechoslovak Mathematical Journal*, 7.
- [14] Ruitenburg, W. (1984). On the period of sequences $(A^n(p))$ in intuitionistic propositional calculus. *The Journal of Symbolic Logic*, 49(3), 892–899.

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